

HW 6 - Q1
To be completed by Wednesday 3/25

1. Use “dataHW6 Problem1.mat” to complete this exercise. The purpose of this problem is to estimate a simple dynamic discrete choice problem. The data contain information on the choices (the Y ’s), characteristics that do not vary over time (the X_1 ’s), and characteristics that do vary over time (the X_{1t} ’s). The context of the problem is a dynamic model of labor supply. Y takes values $\{0, 1, 2, 3\}$ reflecting retirement, flex-time, part-time, or full-time participation. The first column X_1 allows for choice specific constants, while the second column of X_1 is an indicator for gender. X_{1t} is time-varying measure of health.

Both sets of X ’s are individual-level characteristics so the coefficients on these variables need to vary by choice. There is a cost to switching labor supply in the future that is *common across all switches*. The choice in period 0 is given by LY . Once an individual retires (or chooses option 0) then that individual makes no further choices (you can verify this as if $Y_{it} = 0$ then $Y_{it+1} = 0$). The columns of Y and X_{1t} indicate observations over time, ten periods in all (so in period 10 it is a static problem). The only uncertainty from the individual’s perspective is on their error term. In other words, individuals have perfect foresight about their future health. The error structure is Type 1 extreme value.

- (a) Estimate the dynamic discrete choice model including the discount factor.
 - Normalize the flow utility of retirement to zero. Also, since the decision process ends when an individual retires, there is no expected future utility associated with retirement.
 - Once an individual retires, they make no additional contribution to the likelihood.
 - While future health is known with certainty, future choices are not. Thus, there is a time-varying state variable that transitions deterministically based on choice. This affects future utility through the switching cost.
 - True params
 - Coefficients on X ’s (constant, gender, health) for three choices $[-1; .5; 5]$ $[-0.25; 0.75; .75]$ $[-.5; 1; 1]$
 - Switching cost: 0.4 (enters negatively)
 - Discount factor: 0.9
- (b) What variation in the data allows the discount factor to be identified?
- (c) What variation in the data allows the switching cost to be identified?